AJ Barea

Ph.D. Student, Computing and Information Sciences · Rochester Institute of Technology

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Research Interests

Software engineering for federated and distributed machine learning systems; agentic AI with tool use and human oversight; testing, observability, and reproducibility infrastructure for ML research; trustworthy and privacy-preserving learning from human-generated data, including what shared model updates reveal about their source; empirical software engineering on code review; developer experience in AI research tooling.

Education

Doctor of Philosophy, Computing and Information Sciences Aug 2026 to present
Rochester Institute of Technology, Rochester, NY

- NSF AWARE-AI National Research Traineeship (NRT), AI Software track
- Advisor: Dr. Leon Reznik

Master of Science, Software Engineering Dec 2025
Rochester Institute of Technology, Rochester, NY

- GPA: 4.0 / 4.0
- Capstone: *Enhancing Federated Learning Execution Framework*
- Coursework: Software Architecture, Software Quality Assurance, Engineering Cloud Software Systems, Self-Adaptive Systems with Reinforcement Learning, Model-Driven Development, Software Construction, Collaborative Software Development

Bachelor of Science, Computer Engineering May 2022
University of South Florida, Tampa, FL

- Capstone: Private blockchain system for document certification (proof-of-work mining, SHA-256 document signatures, OAuth 2.0 / JWT admin dashboard)

Research Experience

Doctoral Researcher, Federated Agents Aug 2026 to present
Rochester Institute of Technology

- Leading a research direction on coding agents that learn each organization's conventions and pool what generalizes, while each organization decides what may leave
- Designed a pre-registered test of whether learned house style transfers at the project or the organization boundary, written up as a sole-author Stage 1 registered report for MSR 2027
- Built sphragis, an open study harness that collects public Gerrit code review under each host's terms, pseudonymizes it at ingestion, seals the test window by hash, and trains and scores LoRA adapters on Slurm GPU clusters
- Built a source-attribution measurement of whether the shared/private adapter splits in the federated personalization literature reveal the source organization in the half they transmit

Summer Research Scholar May 2026 to Aug 2026
Laboratory for Analytic Sciences (LAS), NC State University · SCADS (Summer Conference on Applied Data Science)

- Selected for a competitive summer research cohort applying data science to analytic sensemaking problems
- Prototyped an agentic sensemaking system on the Claude Agent SDK, tracing an entity's evidence across graph, relational, and text stores into a cited analytic note
- Built a governed MCP orchestration platform for agentic analytic workflows, cutting a measured policy-leak rate from 75% to 0% under a deterministic red-team evaluation battery
- Delivered a motion-first object tracking pipeline for camouflaged targets in video, benchmarked end to end against published state-of-the-art methods
- Instrumented agent and pipeline runs so multi-step reasoning could be inspected, replayed, and evaluated rather than treated as a black box

Graduate Research Assistant May 2025 to present

Laboratory of Data Quality and Intelligent Security (LDQIS), Rochester Institute of Technology · Advisor: Dr. Leon Reznik

- Designed a pytest suite reaching 80%+ coverage for a distributed federated learning framework, converting a research prototype into a dependable, reproducible platform
- Built a React and FastAPI interface that shortened researcher onboarding and improved experiment reproducibility across the group
- Prototyped Byzantine-robust aggregation strategies with Ph.D. collaborators, including Krum, Bulyan, RFA, and Trimmed Mean, plus trust-based federated approaches
- Implemented a dynamic dataset poisoning system enabling adversarial robustness evaluation during federated training
- Integrated transformer-based models for federated learning over medical, legal, and financial text corpora
- Established CI pipelines and cross-platform support (Python 3.9 to 3.11; Windows, macOS, Linux)
- Built the lab's new website (Astro, Tailwind, git-backed CMS) to replace its legacy Flask site

Publications

Refereed Journal

1. Korobeinikov, D., Zatsarenko, R., Chuprov, S., **Barea, A. J.**, & Reznik, L. (2026). InteFL Framework: Optimizing Federated Learning With Metacognition for Application Design and Deployment. *IEEE Intelligent Systems*, 1–10. doi:10.1109/MIS.2026.3658072

Refereed Conference

2. Korobeinikov, D., **Barea, A. J.**, Reznik, L., Zatsarenko, R., Chuprov, S., & Peredo, A. (2026). InteFL: Framework for AI-Assisted Design of Trustworthy and Efficient Federated Learning Applications. *International Joint Conference on Neural Networks (IJCNN), IEEE World Congress on Computational Intelligence*, Maastricht, Netherlands.

Refereed Workshop

3. **Barea, A. J.** (2026). Kourai Khryseai: Transparent Human-on-the-Loop Multi-Agent Software Development. *North-East AI Agents Day 2026 Workshop*, Jane Street, New York, NY. Accepted as poster (sole author). OpenReview: jdD8sORLLY

Technical Report

4. Padilla, J. G., **Barea, A. J.**, Johnson, B. K., & Tarosky, J. H. (2026). makesense: A Governed Offline Interface for Multi-Source Intelligence Sensemaking. In *SCADS 2026 Technical Report*, 7–34. Laboratory for Analytic Sciences, NC State University. LAS release

5. Johnson, B. K., & **Barea, A. J.** (2026). video-motion: Motion-First Segmentation of Camouflaged Objects in Video at Field-Deployable Speed. In *SCADS 2026 Technical Report*, 253–284. Laboratory for Analytic Sciences, NC State University. LAS release

Under Review

6. **Barea, A. J.**, Johnson, B. K., Tarosky, J. H., & Padilla, J. G. (2026). makesense: Governed Agentic Sensemaking over Heterogeneous Data Stores for Intelligence Analysis. Submitted to *Innovative Applications of Artificial Intelligence (IAAI-27)*.

In Preparation

7. **Barea, A. J.** Do Organizations Leave a Learnable House Style in Code Review? Stage 1 registered report, *MSR 2027 Registered Reports* track with *Empirical Software Engineering* (sole author).

Software

8. **Barea, A. J.** (2026). techne: a Claude Code plugin marketplace for repositories, documents, and bibliographies. Zenodo. doi:10.5281/zenodo.23330765

9. **Barea, A. J.** (2026). keryx: gives Claude Code a voice. Zenodo. doi:10.5281/zenodo.23327640

Technical Skills

Machine Learning: PyTorch, Hugging Face Transformers and PEFT (LoRA), scikit-learn, Flower, federated learning, reinforcement learning, Webots

Agentic AI: Claude Agent SDK, Model Context Protocol (MCP), A2A protocol, tool-use orchestration, LLM evaluation

Languages: Python, Java, TypeScript, JavaScript, C++, C, SQL, Rust

Web and Services: FastAPI, React, Vue.js, Spring Boot, Node.js

Infrastructure: Docker, Terraform, GitHub Actions, CI/CD, Slurm HPC clusters, AWS (Lambda, EC2, S3, DynamoDB, API Gateway, CloudFront, Rekognition, Comprehend, Translate)

Observability and QA: pytest, OpenTelemetry, Jaeger, Prometheus, coverage analysis

Collaboration Tools: Git, Jira, Confluence

Selected Research Software

Ariadne · SCADS sensemaking prototype built on the Claude Agent SDK, with MCP tool servers, Neo4j and PostgreSQL knowledge stores, and OpenTelemetry tracing.

Kourai Khryseai · Multi-agent system using the A2A protocol and MCP with real-time speech synthesis, telemetry through OpenTelemetry, Jaeger, and Prometheus. Presented as a poster at NE Agents Day 2026.

Velocity-FL · Federated learning research toolkit with a Rust core and Python bindings (maturin), PyTorch and Hugging Face model support, and an MCP server interface.

Phalanx-FL · Federated LoRA testbed on the current Flower release with OpenTelemetry tracing that follows every round down to each client and records why each failed client failed.

techne · Claude Code plugin marketplace of agent workflows for repository maintenance, adversarial code review, LaTeX and slide builds, and bibliography verification against arXiv and Crossref; includes **keryx**, spoken session replies from a local model and local speech synthesis.

Pharos · Federated personalization testbed with security-labeled synthetic data, measuring what a personal/shared boundary costs and how much privacy leaks past it under secure aggregation and audit policy.

Security Evaluation System · Hybrid evaluation tool combining CLIPS rule-based expert systems with PyTorch models to assess Byzantine fault tolerance in distributed machine learning.

BioRadio EMG-to-MIDI · Real-time pipeline mapping EMG and EEG biosignals to musical output over Lab Streaming Layer. Built at the AWARE-AI Spring Hackathon, where the team won Best AI Model.

Autonomous Robot Navigation (RL) · Q-learning and DQN agents in the Webots simulator with state discretization, reward shaping, replay buffers, and PyTorch hyperparameter management.

Teaching Experience

Teaching Assistant, Software Architecture Aug 2025 to Dec 2025
Rochester Institute of Technology

- Mentored students in design patterns, architectural styles, and quality attributes and tactics
- Led recitation sections and office hours; taught a full class session during instructor absence
- Graded assignments with detailed feedback emphasizing rigorous engineering practice

Presentations and Posters

Talk, *Federated Agents: Coding Agents That Learn Each Company's House Rules Without Its Code Leaving*. Rochester Institute of Technology. October 2026. Slides

Poster #26, *Narrative Interfaces for Multi-Agent Software Development*. CHAI Research Poster Session, RIT Center for Human-aware Artificial Intelligence, Student Innovation Hall, Rochester, NY. April 2026.

Talk, *TraceFL: Debugging Federated Learning Systems*. Software Quality Assurance, Rochester Institute of Technology. Fall 2025.

Tutorial series, applied ML, reinforcement learning, and AI developer tooling. YouTube (@ajbarea). 2024 to present.

Professional Experience

Software Engineer (Contract) Feb 2023 to Aug 2024
Lowe's Home Improvement, Remote · Distribution Management Systems

- Cut daily error logs by 150,000 entries across regional distribution centers through systematic debugging, log analysis, and performance work
- Built a full-stack internal tool for database querying and SQL script generation, reducing support ticket resolution time

- Authored standard operating procedures that reduced new engineer onboarding time by 80%
- Performed Tier 3 root cause analysis and delivered tested, documented production fixes

UX Developer (Intern) Mar 2022 to May 2022
Phiquest, Tampa, FL

- Designed wireframes for web applications with cross-functional teams
- Engineered a blockchain-based document authenticator Chrome extension in Node.js

Honors and Awards

NSF AWARE-AI National Research Traineeship, Rochester Institute of Technology	2026
SCADS Summer Research Scholar, Laboratory for Analytic Sciences (LAS), NC State University	2026
HackerRank Orchestrate 2026 Hackathon · 11th of 1,349 (top 0.8%), solo entry	2026
Best AI Model, AWARE-AI Spring Hackathon, Rochester Institute of Technology · team entry	2026
Certificate of Performance, Phiquest	2022
The National Society of Leadership and Success	2019

Service and Outreach

Science communication: Produce open educational content on AI development tooling, pair programming with AI assistants, and reinforcement learning for a public audience of students and developers (YouTube @ajbarea).

Open source: Maintain public research software and documentation across federated learning, agentic AI, and developer tooling projects (github.com/ajbarea).

Program participation: AWARE-AI Spring Hackathon (2026); AWARE-AI workshops on research storytelling and collaborative research; NSF SaTC-EDU AI4IA training on LLM-assisted cyber incident analysis.